

Substitution: Reading an Integral Backwards Through the Chain Rule

- Problem 1 shows the **substitution table** filled in. Build the same three-row table for every problem after it — the table is the bookkeeping, and most substitution errors are bookkeeping errors.
- For an indefinite integral, convert back to x and add $+C$. For a definite integral, convert the *limits* instead and never return to x .
- Give exact values (fractions, radicals, e) throughout.

1. Evaluate $\int 2x (x^2 + 7)^5 dx$.

inner function	$u = x^2 + 7$
differential	$du = 2x dx$
integral in u	$\int u^5 du$

Finish it, and write your answer in terms of x .

Answer: _____

2. Evaluate $\int x^2 e^{x^3} dx$. Build your own substitution table first.

Answer: _____

3. Consider $\int_0^1 2x (x^2 + 1)^3 dx$ with $u = x^2 + 1$.

(a) Find the value of u at the lower limit $x = 0$.

Answer: _____

(b) Find the value of u at the upper limit $x = 1$.

Answer: _____

(c) Evaluate the integral.

Answer: _____

4. Evaluate $\int \frac{x}{\sqrt{x^2 + 9}} dx$.

Answer: _____

5. Consider $\int_0^{\pi/4} \tan x \sec^2 x dx$ with $u = \tan x$.

(a) Find the value of u at the upper limit $x = \frac{\pi}{4}$.

Answer: _____

(b) Evaluate the integral.

Answer: _____

6. Consider $\int \frac{(\ln x)^3}{x} dx$.

(a) Evaluate it.

Answer: _____

(b) Explain why the $\frac{1}{x}$ in the integrand is exactly what makes this substitution work.

Answer: _____

7. Consider $\int_1^e \frac{\ln x}{x} dx$ with $u = \ln x$.

(a) Find the value of u at the upper limit $x = e$.

Answer: _____

(b) Evaluate the integral.

Answer: _____

8. Consider $\int_0^b x e^{-x^2} dx$ with $u = x^2$.

(a) Evaluate the integral for $b = 2$. Give an exact value.

Answer: _____

(b) In general the integral equals $\frac{1}{2} (1 - e^{-b^2})$. Find its limit as $b \rightarrow \infty$.

Answer: _____

(c) Explain what that limit says about the area under $y = x e^{-x^2}$ for $x \geq 0$.

Answer: _____